

**FACULTY OF AUTOMATION AND COMPUTER SCIENCE
COMPUTER SCIENCE DEPARTMENT**

SUMMARY
of the Master Thesis entitled:

**Authenstickator: a TPM-secured TOTP authenticator app
running on a USB stick**

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1. Requirements

Develop a theoretical framework and a proof-of-concept implementation of an application generating TOTP codes for 2FA authentication securely, running on a USB stick instead of a mobile device. The framework shall be able to:

- Generate TOTP codes
- Store the secrets safely
- Keep the 2FA aspect

The implementation shall be a working application which employs the framework.

2. Proposed solutions

The proposed framework combines three passwords/secrets.

The *disk encryption password* is used by the user to encrypt the whole USB stick. Encryption happens with LUKS2 (on Linux) or BitLocker (on Windows).

The *user password* is required and asked by the application. It is expanded with the PBKDF2 algorithm and used as a secret to define an AES encryption key. AES encryption is used on the storage, i.e., the file containing TOTP secrets. The user password is never saved or logged. Instead, a hash is used to verify login. See the attached diagram for the two main flows of the application.

TPM functionality can be enabled or disabled by the user before first startup. If enabled (on Linux), the TPM chip is used to generate a random number called *TOTP secret*, which is stored in the TPM's NVRAM only. The TPM secret is then used as salt in the AES algorithm. This restricts the usage of the application to one device, hence its optional nature in the framework.

3. Results obtained

The theoretical framework presented in section 2 was proposed, and an application called Authenstickator was developed, which implements it. Source code is available at <https://github.com/broland29/authenstickator> along with 3 ready-to-use releases, for platforms Ubuntu 22.04 LTS, Windows 10 and Ubuntu 24.04 LTS. Involved technologies include Python, HTML, CSS, and JavaScript.



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4. Tests and verifications

The framework's validness was proven by the implementation, which serves as proof of concept. The implementation was manually tested on virtual machines (Ubuntu 22.04, Windows 10, Ubuntu 24.04) and on a Windows 10 machine without Python. Unit tests for business logic were written.

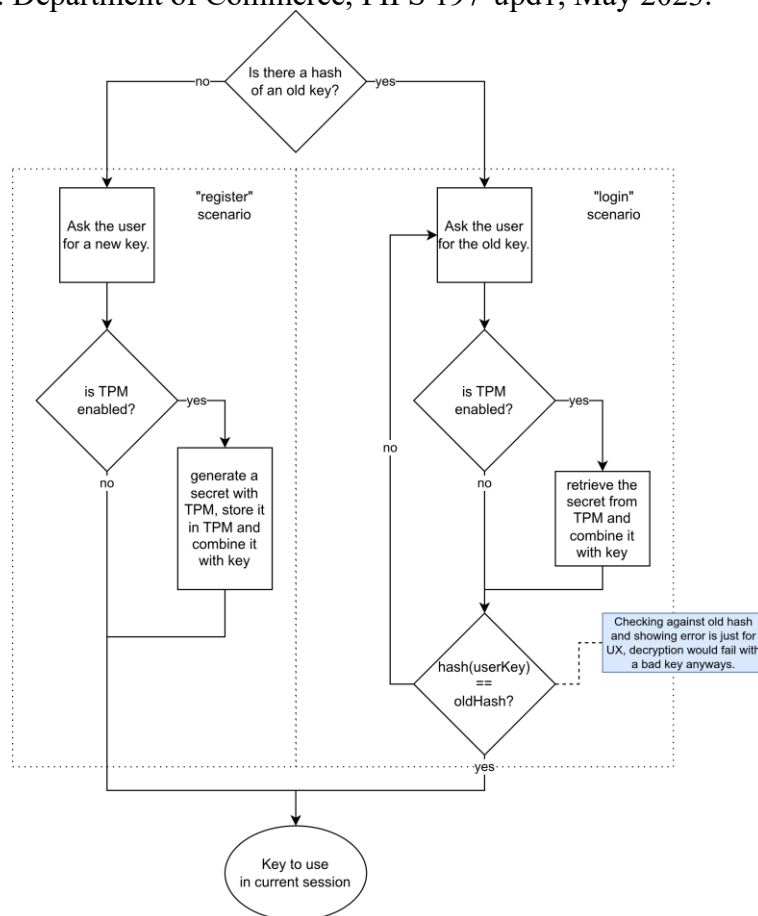
5. Personal contributions

The framework and its implementation. A hands-on proof that running a TOTP application on a USB stick securely is possible.

6. Documentation sources

As defined in the bibliography section of the thesis, including:

- W. Arthur, D. Challener, and K. Goldman, "A Practical Guide to TPM 2.0". Apress, 2015
- D. M'Raihi, S. Machani, M. Pei, and J. Rydell, "TOTP: Time-Based One-Time Password Algorithm," RFC 6238, May 2011
- National Institute of Standards and Technology (NIST), "Advanced Encryption Standard (AES)," U.S. Department of Commerce, FIPS 197-upd1, May 2023.



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